2.10.85

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Problem Statement

Question : Let P be the plane 3x+2y+3z=16 and let S: $\alpha\hat{i} + \beta\hat{j} + \gamma\hat{k}$,where $\alpha + \beta + \gamma = 7$ and the distance of (α, β, γ) from the plane is $2/\sqrt{22}$ .Let $\mathbf{u}, \mathbf{v}, \mathbf{w}$ be three distinct vectors in S such that $|\mathbf{u} - \mathbf{v}| = |\mathbf{v} - \mathbf{w}| = |\mathbf{w} - \mathbf{u}|$. Let V be the volume of the parallelopiped determined by vectors $\mathbf{u}, \mathbf{v}, \mathbf{w}$. Then the value of (80/3)V is

Distance between point and plane

Solution :

$$P: \ \mathbf{n}^\top \mathbf{x} = c, \qquad \mathbf{n} = \begin{pmatrix} 3 \\ 2 \\ 3 \end{pmatrix} \ c = 16. \tag{3.1}$$

The distance of point $\mathbf{P_0} = \begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix}$ from the plane is

$$\mathrm{dist}(\mathbf{P_0}, P) = \frac{|\mathbf{n}^\top \mathbf{P_0} - c|}{\|\mathbf{n}\|} \tag{3.2}$$

Given $\alpha + \beta + \gamma = 7$ and $\mathrm{dist}(\mathbf{P_0}, P) = \frac{2}{\sqrt{22}}$, we have

$$\frac{|3\alpha + 2\beta + 3\gamma - 16|}{\sqrt{22}} = \frac{2}{\sqrt{22}} \implies \mathbf{n}^\top \mathbf{P_0} = 18 \ \text{ or } \ \mathbf{n}^\top \mathbf{P_0} = 14. \tag{3.3}$$

Thus S lies on the intersections

$$\Pi : \ \mathbf{m}^\top \mathbf{x} = 7, \qquad \mathbf{m} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \tag{3.4}$$

with

$$P_+ : \mathbf{n}^\top \mathbf{x} = 18, \qquad P_- : \mathbf{n}^\top \mathbf{x} = 14. \tag{3.5}$$

Each intersection is a line. The direction vector of the line is

$$\mathbf{d} = \mathbf{m} \times \mathbf{n} = \begin{pmatrix} \begin{vmatrix} m_{23} & n_{23} \\ m_{31} & n_{31} \\ m_{12} & n_{12} \end{vmatrix} \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}. \tag{3.6}$$

Equation of line and finding u,v,w

Pick points: For $\Pi \cap P_+$, solve

$$\begin{cases} \mathbf{m}^\top \mathbf{x} = 7, \\ \mathbf{n}^\top \mathbf{x} = 18 \end{cases} \implies \mathbf{p} = \begin{pmatrix} 0 \\ 3 \\ 4 \end{pmatrix} \tag{3.7}$$

So

$$L_+ : \ \mathbf{r} = \mathbf{p} + t\mathbf{d}. \tag{3.8}$$

For $\Pi \cap P_-$, solve

$$\begin{cases} \mathbf{m}^\top \mathbf{x} = 7, \\ \mathbf{n}^\top \mathbf{x} = 14 \end{cases} \implies \mathbf{q} = \begin{pmatrix} 0 \\ 7 \\ 0 \end{pmatrix} \tag{3.9}$$

So

$$L_- : \ \mathbf{r} = \mathbf{q} + t\mathbf{d}. \tag{3.10}$$

Construct equilateral triangle:

Take

$$\mathbf{u} = \mathbf{p} + s\mathbf{d}, \quad \mathbf{v} = \mathbf{p} - s\mathbf{d}, \quad \mathbf{w} = \mathbf{q} + t\mathbf{d}. \tag{3.11}$$

$$\|\mathbf{u} - \mathbf{v}\|^2 = \|\mathbf{u} - \mathbf{w}\|^2 = \|\mathbf{v} - \mathbf{w}\|^2. \tag{3.12}$$

$$\mathbf{u} - \mathbf{v} = 2s\mathbf{d} \;\;\Rightarrow\;\; \|\mathbf{u} - \mathbf{v}\|^2 = 4s^2\|\mathbf{d}\|^2. \tag{3.13}$$

$$\|\mathbf{d}\|^2 = \mathbf{d}^\top\mathbf{d} = 1^2 + 0^2 + (-1)^2 = 2, \|\mathbf{u} - \mathbf{v}\|^2 = 8s^2. \tag{3.14}$$

$$\mathbf{r} = \mathbf{p} - \mathbf{q} = \begin{pmatrix} 0 \\ -4 \\ 4 \end{pmatrix}, \quad \mathbf{u} - \mathbf{w} = \mathbf{r} + (s - t)\mathbf{d}. \tag{3.15}$$

$$\|\mathbf{u} - \mathbf{w}\|^2 = \|\mathbf{r}\|^2 + 2(s - t)(\mathbf{r}^\top\mathbf{d}) + (s - t)^2\|\mathbf{d}\|^2. \tag{3.16}$$

$$\|\mathbf{r}\|^2 = 32, \quad \mathbf{r}^\top \mathbf{d} = -4, \quad \|\mathbf{d}\|^2 = 2. \tag{3.17}$$

$$\|\mathbf{u} - \mathbf{w}\|^2 = 32 - 8(s-t) + 2(s-t)^2. \tag{3.18}$$

$$\mathbf{v} - \mathbf{w} = \mathbf{r} + (-s-t)\mathbf{d} \;\;\Rightarrow\;\; \|\mathbf{v} - \mathbf{w}\|^2 = 32 + 8(s+t) + 2(s+t)^2. \tag{3.19}$$

Equating

$$32 - 8(s-t) + 2(s-t)^2 = 8s^2, \quad 32 + 8(s+t) + 2(s+t)^2 = 8s^2. \tag{3.20}$$

$$16s + 8st = 0 \quad \Rightarrow \quad t = -2,\; s^2 = 4. \tag{3.21}$$

Take $s = 2, t = -2$:

$$\mathbf{u} = \mathbf{p} + 2\mathbf{d} = \begin{pmatrix} 2 \\ 3 \\ 2 \end{pmatrix}, \quad \mathbf{v} = \mathbf{p} - 2\mathbf{d} = \begin{pmatrix} -2 \\ 3 \\ 6 \end{pmatrix}, \quad \mathbf{w} = \mathbf{q} - 2\mathbf{d} = \begin{pmatrix} -2 \\ 7 \\ 2 \end{pmatrix}.$$

Gram matrix and Volume

Volume of parallelepiped:

Gram matrix G (matrix of pairwise inner products):

$$G = \begin{pmatrix} \|\mathbf{u}\|^2 & \mathbf{u}^\top\mathbf{v} & \mathbf{u}^\top\mathbf{w} \\ \mathbf{v}^\top\mathbf{u} & \|\mathbf{v}\|^2 & \mathbf{v}^\top\mathbf{w} \\ \mathbf{w}^\top\mathbf{u} & \mathbf{w}^\top\mathbf{v} & \|\mathbf{w}\|^2 \end{pmatrix} = \begin{pmatrix} 17 & 17 & 21 \\ 17 & 49 & 37 \\ 21 & 37 & 57 \end{pmatrix}, \tag{3.23}$$

Determinant of the Gram matrix (gives the squared volume):

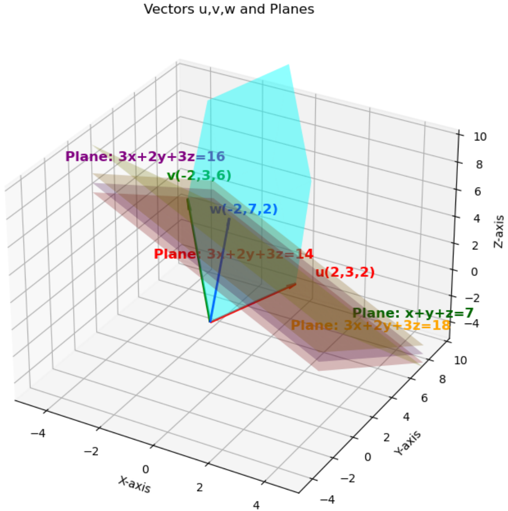
$$\det G = det \begin{pmatrix} 17 & 17 & 21 \\ 17 & 49 & 37 \\ 21 & 37 & 57 \end{pmatrix} = 12544 \tag{3.24}$$

Hence the volume itself is

$$V = \sqrt{\det G} = \sqrt{12544} = 112. \tag{3.25}$$

$$\frac{80V}{3} = \frac{80x112}{3} = \frac{8960}{3}. \tag{3.26}$$

Plots



Vectors u,v,w and Planes
Plane: 3x+2y+3z=16
v(-2,3,6)
w(-2,7,2)
Plane: 3x+2y+3z=14
u(2,3,2)
Plane: x+y+z=7
Plane: 3x+2y+3z=18
X-axis
Y-axis
Z-axis

C Code

```
#include <stdio.h>
#include <math.h>

// Function to compute determinant of 3x3 matrix
double determinant(double a[3][3]) {
    return a[0][0]*(a[1][1]*a[2][2] - a[1][2]*a[2][1])
        - a[0][1]*(a[1][0]*a[2][2] - a[1][2]*a[2][0])
        + a[0][2]*(a[1][0]*a[2][1] - a[1][1]*a[2][0]);}
int main() {
    // Two planes:
    // Plane P: 3x+2y+3z=16
    // Plane S: x+y+z=7
    // Solve intersection line between P and S
    // Choose basis for S: let
    // a = (1,-1,0), b = (1,0,-1)  both satisfy x+y+z=0
```

```
// So general point on S: (7,0,0) + s*a + t*b
// Constraint from P: 3x+2y+3z = 14 or 18 (distance condition)

// Pick "14" first.
// Solve quickly: Intersection point
double u[3] = {2,3,2};
double v[3] = {-2,3,6};
double w[3] = {-2,7,2};

// Put in matrix
double M[3][3] = {
    {u[0], v[0], w[0]},
    {u[1], v[1], w[1]},
    {u[2], v[2], w[2]}};
// Determinant
double det = determinant(M);
double V = fabs(det);
double result = (80.0/3.0) * V;
```

```
    printf("Chosen vectors:\n");
    printf("u = (%.2lf, %.2lf, %.2lf)\n", u[0], u[1], u[2]);
    printf("v = (%.2lf, %.2lf, %.2lf)\n", v[0], v[1], v[2]);
    printf("w = (%.2lf, %.2lf, %.2lf)\n", w[0], w[1], w[2]);
    printf("Determinant = %.2lf\n", det);
    printf("Volume V = %.2lf\n", V);
    printf("Final (80/3)*V = %.2lf\n", result);

    return 0;
}
```

Python : call_c.py

```
import ctypes
import os

# Load shared library
lib = ctypes.CDLL(os.path.abspath("./libvolume.so"))

# Set return types
lib.get_determinant.restype = ctypes.c_double
lib.get_volume.restype = ctypes.c_double
lib.compute_volume.restype = ctypes.c_double
lib.get_u.restype = ctypes.c_double
lib.get_v.restype = ctypes.c_double
lib.get_w.restype = ctypes.c_double

# Fetch vectors u, v, w
u = [lib.get_u(i) for i in range(3)]
v = [lib.get_v(i) for i in range(3)]
```

```
w = [lib.get_w(i) for i in range(3)]

# Fetch determinant, volume, and final result
det = lib.get_determinant()
V = lib.get_volume()
result = lib.compute_volume()

# Print everything
print("Vector u =", u)
print("Vector v =", v)
print("Vector w =", w)
print("Determinant =", det)
print("Volume V =", V)
print("Final (80/3)*V =", result)
```

Python Code for Plotting

```
import numpy as np
import matplotlib.pyplot as plt
from mpl_toolkits.mplot3d.art3d import Poly3DCollection

# Define vectors
u = np.array([2, 3, 2])
v = np.array([-2, 3, 6])
w = np.array([-2, 7, 2])
# Origin
O = np.array([0, 0, 0])
# Create 3D plot
fig = plt.figure(figsize=(10, 8))
ax = fig.add_subplot(111, projection='3d')
# ---- Plot Vectors u, v, w ----
def plot_vector(vec, color, label, offset):
    ax.quiver(O[0], O[1], O[2],
              vec[0], vec[1], vec[2],
```

```
                  color=color, arrow_length_ratio=0.1, linewidth=2)
    ax.text(vec[0]+offset[0], vec[1]+offset[1], vec[2]+offset[2],
                label, fontsize=12, color=color, weight="bold")

plot_vector(u, "red", "u(2,3,2)", offset=(0.5, 0.5, 0.5))
plot_vector(v, "green", "v(-2,3,6)", offset=(-1, 0.5, 0.5))
plot_vector(w, "blue", "w(-2,7,2)", offset=(-1, 0.5, -0.5))

# ---- Plot parallelepiped ----
vertices = np.array([
    O, u, v, w,
    u+v, v+w, w+u, u+v+w
])
faces = [
    [O, u, u+v, v],
    [O, v, v+w, w],
    [O, u, u+w, w],
    [u, u+v, u+v+w, u+w],
    [v, u+v, u+v+w, v+w],
```

```
    [w, v+w, u+v+w, u+w]]
ax.add_collection3d(Poly3DCollection(faces, alpha=0.25, facecolor="cyan
    "))

# ----- Planes -----
xx, yy = np.meshgrid(range(-3, 6), range(-3, 9))

# Plane 1: 3x+2y+3z = 18
zz1 = (18 - 3*xx - 2*yy) / 3
ax.plot_surface(xx, yy, zz1, alpha=0.25, color="orange")
ax.text(4, -3, (18 - 3*4 - 2*(-3)) / 3,
        "Plane: 3x+2y+3z=18", color="orange", fontsize=12, weight="
            bold")
# Plane 2: 3x+2y+3z = 14 (label at bottom of plane)
zz2 = (14 - 3*xx - 2*yy) / 3
ax.plot_surface(xx, yy, zz2, alpha=0.25, color="red")
ax.text(0, -5, (14 - 3*0 - 2*(-5)) / 3,
        "Plane: 3x+2y+3z=14", color="red", fontsize=12, weight="
            bold")
```

```
# Plane 3: x+y+z = 7 (plane yellow, label dark green)
zz3 = 7 - xx - yy
ax.plot_surface(xx, yy, zz3, alpha=0.25, color="yellow")
ax.text(3, 6, 7 - 3 - 6,
        "Plane: x+y+z=7", color="darkgreen", fontsize=12, weight="
            bold")

# Plane 4: 3x+2y+3z = 16 (mid-plane)
zz4 = (16 - 3*xx - 2*yy) / 3
ax.plot_surface(xx, yy, zz4, alpha=0.25, color="purple")
ax.text(-4, -3, (16 - 3*(-4) - 2*(-3)) / 3,
        "Plane: 3x+2y+3z=16", color="purple", fontsize=12, weight="
            bold")

# ---- Axes settings ----
ax.set_xlim(-5, 5)
ax.set_ylim(-5, 10)
ax.set_zlim(-5, 10)
```

```
ax.set_xlabel("X-axis")
ax.set_ylabel("Y-axis")
ax.set_zlabel("Z-axis")
ax.set_title("Vectors u,v,w and Planes")

plt.show()
```